

Class Programming Homework: Search Algorithm

8-Puzzle Problem — DFS and A* Search

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1. Introduction

This report presents the implementation and results of search algorithms applied to the 8-puzzle problem. The student ID is **225040015** (odd number), so **Depth-First Search (DFS)** is the required primary algorithm, together with **A* Search** using the Manhattan distance heuristic. The digit 0 represents the blank tile.

2. Problem Representation

The 8-puzzle board is encoded as a 9-character string in row-major order, where 0 is the blank tile:

```
String index: 0 1 2 3 4 5 6 7 8
Grid layout:  [0] [1] [2]
               [3] [4] [5]
               [6] [7] [8]

Example: '283164705' =>  2 8 3
                        1 6 4
                        7 0 5   (0 = blank tile)
```

Parity is determined by counting *inversions* in the tile sequence (ignoring 0). An inversion is a pair (i, j) where $i < j$ but $\text{tile}[i] > \text{tile}[j]$. Two states are mutually reachable if and only if their inversion counts share the same parity (both odd or both even).

3. Algorithms

3.1 Depth-First Search (DFS)

Uses an explicit stack with a **global visited set** to prevent revisiting states, and a **depth limit of 50** to avoid infinite traversal. DFS does *not* guarantee the shortest (optimal) path.

3.2 A* Search

Uses a min-heap ordered by $f(n) = g(n) + h(n)$, where $g(n)$ is the path cost from the start and $h(n)$ is the **Manhattan distance heuristic**:

$$h(n) = \sum_{\text{tile} \neq 0} (|\text{row}_{\text{current}} - \text{row}_{\text{goal}}| + |\text{col}_{\text{current}} - \text{col}_{\text{goal}}|)$$

This heuristic is *admissible* and *consistent*, guaranteeing the optimal solution.

4. Task A1 — DFS on Fixed States

Start State: 283164705 Goal State: 123804765

2 8 3		1 2 3
1 6 4	==>	8 0 4
7 0 5		7 6 5

DFS Result:

Algorithm : DFS (Depth-First Search)
 Nodes expanded : 101,085
 Number of moves: 5

Solution Path:

Step	Row 1	Row 2	Row 3	State String
Start	2 8 3	1 6 4	7 0 5	283164705
1	2 8 3	1 0 4	7 6 5	283104765
2	2 0 3	1 8 4	7 6 5	203184765
3	0 2 3	1 8 4	7 6 5	023184765
4	1 2 3	0 8 4	7 6 5	123084765
lightblue Goal	1 2 3	8 0 4	7 6 5	123804765

5. Task B1 — A* on Fixed States

Same start and goal states as Task A1.

A* Result:

Algorithm : A* (Manhattan Distance Heuristic)
 Nodes expanded : 6
 Number of moves: 5

Solution Path (identical to DFS — both find the 5-move optimum):

Step	Row 1	Row 2	Row 3	State String
Start	2 8 3	1 6 4	7 0 5	283164705
1	2 8 3	1 0 4	7 6 5	283104765
2	2 0 3	1 8 4	7 6 5	203184765
3	0 2 3	1 8 4	7 6 5	023184765
4	1 2 3	0 8 4	7 6 5	123084765
lightblue Goal	1 2 3	8 0 4	7 6 5	123804765

Key observation: A* expands only **6 nodes** versus DFS's **101,085** — a **16,847×** improvement — while finding the same 5-move optimal path.

6. Task C1 — Parity Test

The first two tiles (2 and 8) are swapped in the start state:

```
Original start : 283164705 (inversions: 11 -- ODD)
Swapped start  : 823164705 (inversions: 12 -- EVEN)
Goal state     : 123804765 (inversions: 7 -- ODD)

same_parity(original, goal) = True  => Solvable
same_parity(swapped, goal)  = False => Unsolvable
```

Running DFS on the swapped start exhausts all reachable states and confirms **no solution exists**. This verifies the parity invariant: swapping any two tiles changes the inversion count parity, making the goal state unreachable.

7. Tasks A2 & B2 — Random States (Seed = 225040015)

Random start and goal states are generated using `random.Random(225040015)` (student ID as seed):

```
Random seed      : 225040015 (student ID)
Random start     : 241570638 (inversions: 7 -- ODD)
Random goal      : 740138562 (inversions: 15 -- ODD)
Same parity      : True -- no state adjustment needed
```

Random Start State (241570638):	Random Goal State (740138562):
2 4 1	7 4 0
5 7 0	1 3 8
6 3 8	5 6 2

7.1 Task A2 — DFS on Random States

```
Algorithm       : DFS
Nodes expanded  : 107,457
Number of moves : 41
```

Solution Path (DFS, 41 moves):

Step	Row 1	Row 2	Row 3	State
lightblue Start	2 4 1	5 7 0	6 3 8	241570638
1	2 4 0	5 7 1	6 3 8	240571638
2	2 0 4	5 7 1	6 3 8	204571638
3	2 7 4	5 0 1	6 3 8	274501638
4	2 7 4	5 3 1	6 0 8	274531608
5	2 7 4	5 3 1	0 6 8	274531068
6	2 7 4	0 3 1	5 6 8	274031568
7	0 7 4	2 3 1	5 6 8	074231568
8	7 0 4	2 3 1	5 6 8	704231568
9	7 3 4	2 0 1	5 6 8	734201568
10	7 3 4	2 1 0	5 6 8	734210568
11	7 3 4	2 1 8	5 6 0	734218560
12	7 3 4	2 1 8	5 0 6	734218506
13	7 3 4	2 0 8	5 1 6	734208516
14	7 0 4	2 3 8	5 1 6	704238516
15	0 7 4	2 3 8	5 1 6	074238516
16	2 7 4	0 3 8	5 1 6	274038516
17	2 7 4	3 0 8	5 1 6	274308516
18	2 0 4	3 7 8	5 1 6	204378516
19	0 2 4	3 7 8	5 1 6	024378516
20	3 2 4	0 7 8	5 1 6	324078516
21	3 2 4	5 7 8	0 1 6	324578016
22	3 2 4	5 7 8	1 0 6	324578106
23	3 2 4	5 0 8	1 7 6	324508176
24	3 0 4	5 2 8	1 7 6	304528176
25	3 4 0	5 2 8	1 7 6	340528176
26	3 4 8	5 2 0	1 7 6	348520176
27	3 4 8	5 0 2	1 7 6	348502176
28	3 4 8	5 7 2	1 0 6	348572106
29	3 4 8	5 7 2	0 1 6	348572016
30	3 4 8	0 7 2	5 1 6	348072516
31	3 4 8	7 0 2	5 1 6	348702516
32	3 4 8	7 1 2	5 0 6	348712506
33	3 4 8	7 1 2	5 6 0	348712560
34	3 4 8	7 1 0	5 6 2	348710562
35	3 4 0	7 1 8	5 6 2	340718562
36	3 0 4	7 1 8	5 6 2	304718562
37	0 3 4	7 1 8	5 6 2	034718562
38	7 3 4	0 1 8	5 6 2	734018562
39	7 3 4	1 0 8	5 6 2	734108562
40	7 0 4	1 3 8	5 6 2	704138562
lightblue Goal	7 4 0	1 3 8	5 6 2	740138562

7.2 Task B2 — A* on Random States

Algorithm : A* (Manhattan Distance Heuristic)
Nodes expanded : 612
Number of moves: 21

Solution Path (A*, 21 moves — optimal):

Step	Row 1	Row 2	Row 3	State
lightblue Start	2 4 1	5 7 0	6 3 8	241570638
1	2 4 0	5 7 1	6 3 8	240571638
2	2 0 4	5 7 1	6 3 8	204571638
3	2 7 4	5 0 1	6 3 8	274501638
4	2 7 4	5 3 1	6 0 8	274531608
5	2 7 4	5 3 1	0 6 8	274531068
6	2 7 4	0 3 1	5 6 8	274031568
7	0 7 4	2 3 1	5 6 8	074231568
8	7 0 4	2 3 1	5 6 8	704231568
9	7 3 4	2 0 1	5 6 8	734201568
10	7 3 4	0 2 1	5 6 8	734021568
11	7 3 4	5 2 1	0 6 8	734521068
12	7 3 4	5 2 1	6 0 8	734521608
13	7 3 4	5 0 1	6 2 8	734501628
14	7 3 4	5 1 0	6 2 8	734510628
15	7 3 4	5 1 8	6 2 0	734518620
16	7 3 4	5 1 8	6 0 2	734518602
17	7 3 4	5 1 8	0 6 2	734518062
18	7 3 4	0 1 8	5 6 2	734018562
19	7 3 4	1 0 8	5 6 2	734108562
20	7 0 4	1 3 8	5 6 2	704138562
lightblue Goal	7 4 0	1 3 8	5 6 2	740138562

8. Summary of Results

Task	Algorithm	Start State	Goal State	Result
A1	DFS	283164705	123804765	5 moves (101,085 nodes)
B1	A*	283164705	123804765	5 moves (6 nodes)
C1	DFS (Parity)	823164705	123804765	No solution — diff. parity
A2	DFS	241570638	740138562	41 moves (107,457 nodes)
B2	A*	241570638	740138562	21 moves (612 nodes)

9. Analysis and Conclusion

Efficiency. A* dramatically outperforms DFS in terms of nodes expanded. For the fixed puzzle, DFS expanded **101,085** nodes while A* expanded only **6**. For the random puzzle: **107,457** versus **612**. The Manhattan distance heuristic effectively prunes the search space.

Solution Quality. A* guarantees the optimal (shortest) path. DFS found 5 moves for the fixed puzzle — coincidentally matching the optimal — but required **41 moves** for the random puzzle, nearly double A*'s optimal **21 moves**.

Parity. The parity test confirms a fundamental invariant of the 8-puzzle: swapping any two tiles changes the inversion count by an odd number, flipping parity. DFS correctly exhausted all reachable states with no solution found when parity differs.

Conclusion. A* with the Manhattan distance heuristic is highly efficient and guaran-

tees optimal solutions for the 8-puzzle. DFS, while complete with visited-set tracking and depth limiting, is neither optimal nor computationally efficient for more complex instances.

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